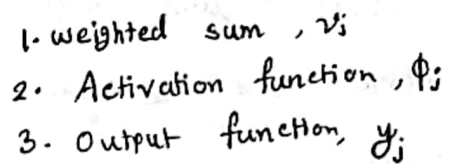
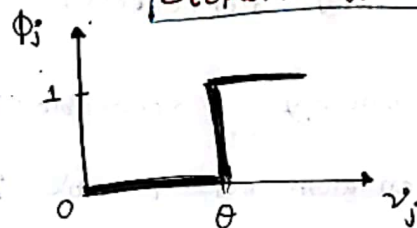


scalar format $ax + by = c$
linear equation 3-
2ଟା.



Stepwise function



Φ_j vs $\mathcal{V}_j$ graph

express করা যায়,

Sign function



The graph shows a step function on a Cartesian coordinate system. The horizontal axis is the x-axis, and the vertical axis is the y-axis. The function is defined as $y = -1$ for $x < 0$ and $y = 1$ for $x > 0$. There is a jump discontinuity at $x = 0$, where the function is not defined (or has multiple values). The y-axis has tick marks at 1 and -1. The x-axis has a tick mark at 0.

$$\phi_j = \begin{cases} 1 & ; v_j \geq \theta \\ 0 & ; v_j < \theta. \end{cases}$$

if dataset doesn't contain

$$\Phi_j = \begin{cases} +1 & ; v_j \geq 0 \\ -1 & ; v_j < 0 \end{cases}$$

$\phi_j = \begin{cases} -1 & ; & v_j \leq 0 \end{cases}$

* ऐसे 2 टो activation function representation ϕ_j theoretical
 concept ϕ_j (उदा), ϕ_j real world ϕ_j applicable 21,
 एव ϕ_j pragmatic boolean function ϕ_j use 22,

real world এর সবকিছু boolean দিয়ে express করা সম্ভব নয়।

Linear separability

একটি dataset এ binary data থাকলে তাকে linearly separable বলে।
একটি একটি line দিয়ে আলাদা করা যায় কিনা।
এ plane এর 2D ক্ষেত্রে, $\rightarrow$ known as data linear separability.

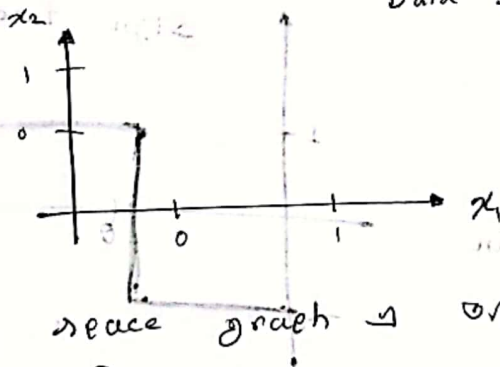
- আমরা data to linear separability applicable করতে পারলে single perceptron দিয়ে representation করা যায়।
- আমরা data linearly separable না হলে one single perceptron দিয়ে representation not possible; তাহলে তখন MLP লাগে।

$\rightarrow$ XOR (example)

both XOR, XNOR gate linearly separable নয়।

Feature Map:

Variable বা feature হিসাবে ডাটা axis এ embed করে দিবে।
Data Space Graph



- data space graph এ আমরা feature map identify করতে try করব।



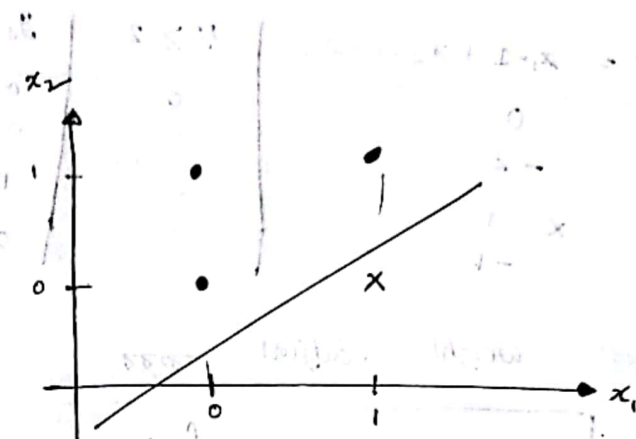
$$x_1 \bar{x}_2 = x_1 \text{ AND NOT } x_2$$

$$x_1 \oplus x_2 = x_1 \bar{x}_2 + x_2 \bar{x}_1$$

is $x_1 \bar{x}_2$ linearly separable?

⇒

x_1	x_2	$\bar{x}_2$	$x_1 \bar{x}_2$	compute
0	0	1	0	
0	1	0	0	
1	0	1	1	
1	1	0	0	



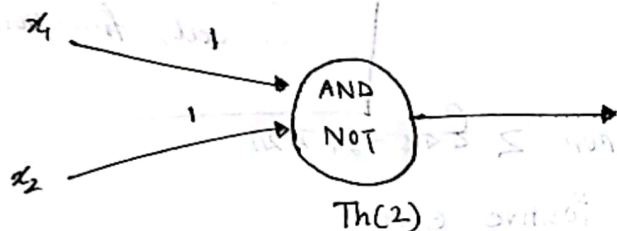
∴ linearly separable.

অন্যভাবে তুলেলে এই line টি identify করা; perceptron এই line টি

produce করা হয় করতে.

∴ dataset টাটার একটা line দিয়ে 2 ভাগ করা যায়.

possible হবে ∴ linearly separable



$v_j = x_1 \cdot 1 + x_2 \cdot 1$	$v_j \geq 2$	y_j
0	0	0
1	0	0
1	0	0 [perceptron isn't correct]
2	1	1 [perceptron isn't correct]

এই ক্ষেত্রে made error

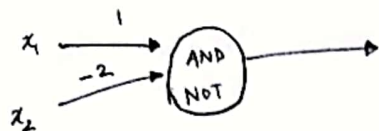
equation একটা variable ২ টি (w, θ) অন্যভাবে এই ২ টি variable এর মধ্যে either threshold change করা or weight change করা not both.

• "Threshold change করা or weight change করা তা depends on model (model builder এর উপর depend করে)"

weight positive ৩ হবে মানে; weight negative ৩ হবে মানে

if we multiply one thing with a fractional part; it reduces.

ଓଟର error ଠିକ୍, but error amount ଛାଡ଼େ.



$$y_j = x_1 \cdot 1 + x_2 \cdot (-2)$$

	$y_j \geq 2$	y_j
0	0	0
-2	0	0
1	0	0
-1	0	0

0 [Perceptron isn't correct]

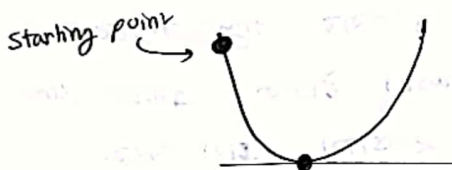
non zero ଓଟର weight update ଥାଏ.

ଓଟର weight adjust ଥାଏ.

$$w_{new} = w_{old} + \text{factor} * \text{error} * I_j$$

factor can be integer or float as well, fraction part.

factor ମଧ୍ୟ error minimize କରାଏ try କର.



error $\geq$ ଥିବା ସମୟରେ:

- Positive error
- Negative error

error positive ଥାଏ, $(y_j - y)$

$$(y_j - x_1 x_2)$$

ଓଟର weight ଛାଡ଼ାଯାଏ.

y_j ଯଦି ଥାଏ ତାହା weight ଠିକ୍ କରାଯାଏ push କରାଯାଏ ତାହା y_j certain value ଦିଆଯାଏ ତାହା ଠିକ୍.

error negative ଥାଏ,

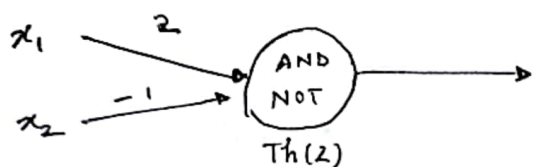
computed value desired value ଠିକ୍ କର.

ମିଥ୍ୟା $\Rightarrow$ weight ଛାଡ଼ାଯାଏ.

- Unlimited time weight update કરવા સમજી સમજી difference (error) 0 થી રહે, Computer ને કોઈ logic ના criteria ની રીતે error ના loss રહેતી રહેતી weight અમલ થાય.

• weight ના સમજી threshold ના relation થી જાણી શકાય, mark કરવા જોઈએ જોઈએ error બાંધી, નો કોઈ કોઈ factor નીમી બાંધી શકાય શકાય error minimize થાય

- mark કરવા જોઈએ જોઈએ error બાંધી, નો કોઈ કોઈ factor નીમી બાંધી શકાય શકાય error minimize થાય
- early stopping criteria મારી computation



$y_i = 2 \cdot x_1 + (-1) \cdot x_2$	$y_i > 2$	y_i
0	0	0
-1	0	0
2	1	1
1	0	0

Computer એટલે કે vector નીમી,

- Tensor is a representation of a vector.

$\therefore x_1, \bar{x}_2$ is linearly separable and perceptron oriented.

$$\left. \begin{aligned} z_1 &= x_1 \bar{x}_2 \\ z_2 &= x_2 \bar{x}_1 \end{aligned} \right\} \text{ મારી ઉપર bar તરફ ઉપર weight નો -1 મિલકે,}$$

- Computer scalar ને કે વેક્ટર નો કે, કે vector ને કે વેક્ટર નો કે,

- Threshold નો value નો કે જે કે સમજી identical weight નીમી or gate.

Vector

↳ matrix

↳ row vector

↳ column vector

Perceptron નો

learning procedure.

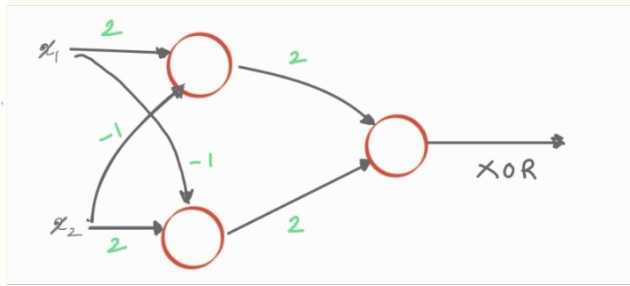


Figure: MLP

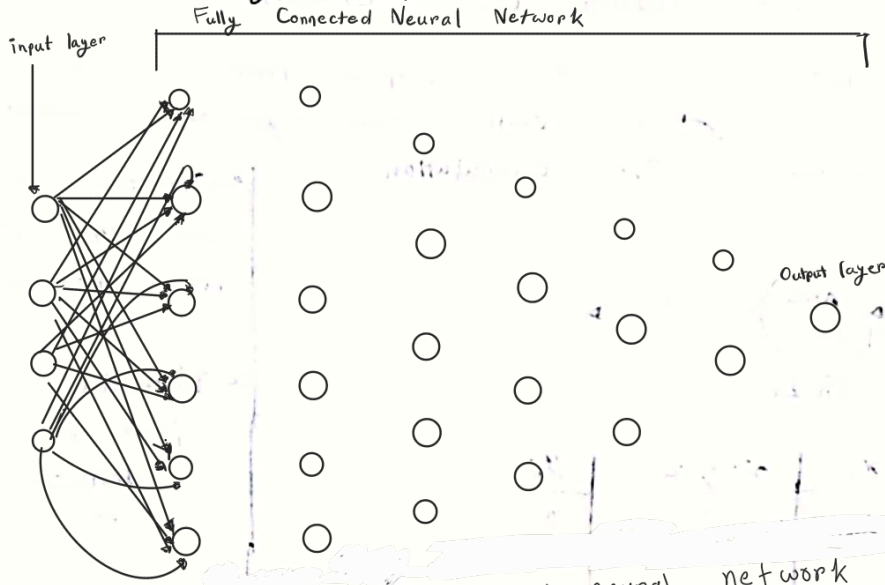


Figure: fully connected neural network

"Fully connected neural network" means all the layers are connected with the previous layers.

How many variables are there in neural network?
 ⇒ Depends on fully connected or not
 fully connected neural network has variable 3.

- ① number of weights aka number of connections
- ② Threshold
- ③ Bias

First we need to identify if the given neural network is fully connected or not.

ଆମର ସମସ୍ତ business ପାଇଁ dataset ଗଠନ କରିବା, dataset ପାଇଁ ଅନେକ style ଏବଂ ଏହାକୁ nature ଏବଂ ଅନ୍ୟ ଭିନ୍ନ ଭିନ୍ନ ନିଆଁ ଏହି nature ଏବଂ ଏହି biased ହେଉଛି, ତେଣୁ, randomize କରି ଉଠାଏ

$$y = mx + (c)_{\text{bias}}$$

"ଆମ Variable ସଂଖ୍ୟା ବଢ଼ାଏ neural network ପାଇଁ learning କରିବାକୁ easy."

number of variables $\uparrow$ number of parameter $\uparrow$ model complexity $\uparrow$ inference time $\uparrow$

- Per neuron ଏକ threshold ଏବଂ bias ଥାଏ.
- input layer x ମିଳିଥାଏ ଯାହାକୁ ନିଆଁ ନିଆଁ feature ଉଠାଏ ତେଣୁ then ଉଠାଏ neuron x ନିଆଁ ନିଆଁ ଗଠନ feature extract କରି then neuron.

ଆମର fully connected neural network,

- Threshold = 27
- Bias = 27.
- number of weights = 130

ଏହା auto extract ହେଉଛି ଏବଂ deep learning ଏହା ଗଠନ କରେ.

2nd layer ଏକ 6 ନୋଡ୍ ଥାଏ ଏବଂ ତାହା ତାହା ତାହା ତାହା ତାହା ତାହା 4 ନୋଡ୍.

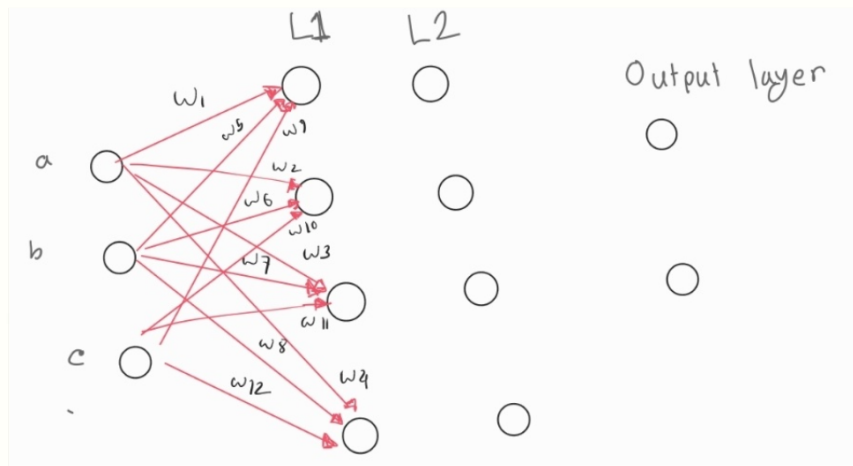
$$(4 \times 6) + (6 \times 6) + (6 \times 5) + (5 \times 4) + (4 \times 3) + (3 \times 2) + (2 \times 1) = 130.$$

ଫିରସ୍ତ layer

$$(w_1, w_2, w_3, \dots, w_{130})$$

$$\therefore \text{Total number of variables} = 130 + 27 + 27 = 184.$$

- Our primary target is to reduce the variable, so that, computation head $\approx$ 0000,



Find the total number of variable for this fully connect neural network

$\Rightarrow$ Input node 3 or.

neurons 10 or ; $\therefore$ threshold = 10 ; bias = 10;

$$\therefore \text{number of weights} = (4 \times 3) + (4 \times 4) + (2 \times 4) \\ = 12 + 16 + 8 \\ = 36$$

$$\therefore \text{Total number of parameters} = 10 + 10 + 36 = 56$$

layerwise scalar representation

L1 (hidden layer 1) or 10 or neuron to z_1 or z_2 denote $\approx$

	1	2	3	4
a	w_1	w_2	w_3	w_4
b	w_5	w_6	w_7	w_8
c	w_9	w_{10}	w_{11}	w_{12}

input
↓
a
b
c

1	2	3	4	neurons
w_1	w_2	w_3	w_4	↓
w_5	w_6	w_7	w_8	
w_9	w_{10}	w_{11}	w_{12}	

$$Z_1 = w_1 a + w_5 b + w_9 c \quad \text{Scalar rep of 1st neuron}$$

$$Z_2 = w_2 a + w_6 b + w_{10} c \quad \text{Scalar rep of 2nd neuron}$$

$$Z_3 = w_3 a + w_7 b + w_{11} c$$

$$Z_4 = w_4 a + w_8 b + w_{12} c$$

• 1st neuron a connection with output.

• 1st row of weighted sum for each column value 1st column output is.

उत्तर scalar representation को vector representation में लिखें,

$$\begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \\ Z_4 \end{bmatrix}_{(4 \times 1)} = \begin{bmatrix} w_1 & w_5 & w_9 \\ w_2 & w_6 & w_{10} \\ w_3 & w_7 & w_{11} \\ w_4 & w_8 & w_{12} \end{bmatrix}_{(4 \times 3)} \begin{bmatrix} a \\ b \\ c \end{bmatrix}_{(3 \times 1)} + \begin{bmatrix} \\ \\ \\ \end{bmatrix}_{(4 \times 1)} \text{ bias}$$

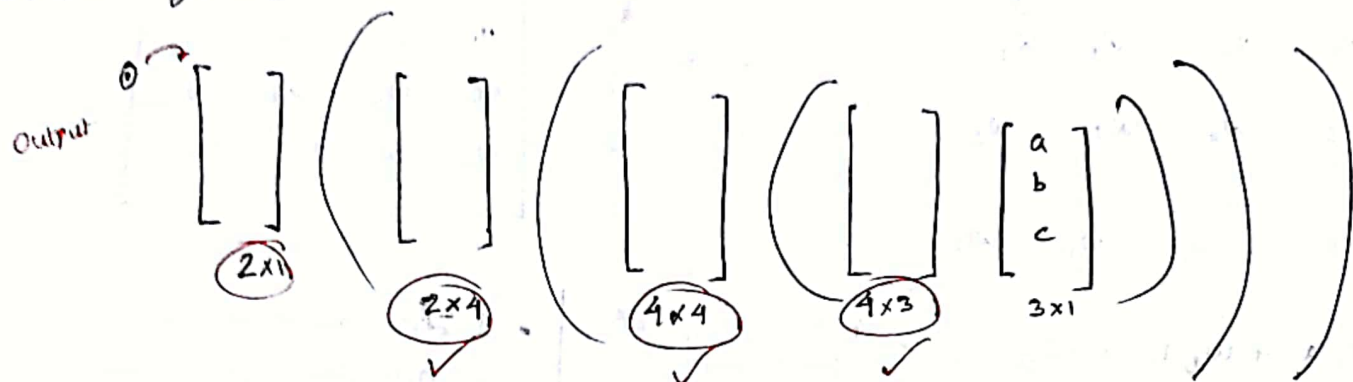
Matrix or vector form.

$$Z = W^T \cdot X + B$$

Transpose of original weight matrix.

$$f(Z) = \begin{bmatrix} f(Z_1) \\ f(Z_2) \\ f(Z_3) \\ f(Z_4) \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \quad \text{L2 to output input hidden layer}$$

Output layer 3.



✓ ডিসকালার সহি sum কবি তাহলে 36 মনে- which is the number of weights.

weight সহি precomputed শরত. তাহলে a, b, c হকসু/ মাত্রে- output মনে.

→ pre trained models.

এক সহি fully connected neural network 'গি' হয়- non fully connected neural network হক matrix representation 'গি' হয়? ওহা কিছু কিছু ডায়গনাল 0 আসক, কিছু

0 সহি represent হক Sparse matrix হয় মাত মনে.

- Scalar matrix
 - Orthogonal matrix
- important terms.
basic concept গিয়াক better.

matrix representation এ ওহা সহি মাত variable control

এ শরত.